

KIDS WHO CHANGED THE WORLD

True Stories of Young Inventors

Ages 8-14

Written by Daniel Tesfamariam

Science Story Series - Book 9

You are NEVER too young to change the world!

These kids proved it.

KIDS WHO CHANGED THE WORLD

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For ages 8-14. Inspiring true stories of young inventors and scientists.

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Chapter 1: Louis Braille

Age 15 - Invented Braille Reading System

Louis Braille lost his sight in an accident when he was only 3 years old.

In 1800s France, blind people had almost no way to read or write.

At school for the blind, Louis was frustrated. The raised-letter books were clumsy.

At age 15, he invented a system of raised dots that could represent any letter!

Using just 6 dots in different patterns, he created an entire alphabet for the blind.

Today, millions of blind people worldwide read using Braille's brilliant system.

THE SCIENCE: HOW TOUCH AND NERVES WORK

Your fingertips have about 3,000 touch receptors per square centimeter!

These receptors detect pressure, vibration, and texture with incredible sensitivity.

Nerve signals from your fingertips travel to your brain at up to 250 mph!

Your brain can distinguish bumps as small as 0.04mm - smaller than a grain of sand!

Blind people often develop even more sensitive touch as their brain adapts.

SCIENCE FACT BOX

- * Braille uses patterns of 1-6 raised dots that represent letters, numbers, and punctuation.
- * Experienced Braille readers can read up to 200 words per minute by touch!
- * Braille is used in over 130 languages worldwide today.

TRY THIS!

Close your eyes and try to identify 5 objects by touch alone. How sensitive are your fingers?

THINK ABOUT IT:

How does losing one sense (like sight) make other senses (like touch) stronger?

Chapter 2: Philo Farnsworth

Age 14 - Conceived the Idea for Television

In 1920, 14-year-old Philo was plowing a potato field in Idaho.

Looking at the parallel rows of soil, he had a brilliant idea!

'What if you could capture an image line by line, like rows in a field?'

He sketched his idea for his high school teacher: electronic television!

By age 21, he built the first fully electronic TV and transmitted an image.

His idea from a potato field changed how the entire world receives information!

THE SCIENCE: HOW SCREENS CREATE IMAGES

TV screens create images using thousands of tiny pixels (picture elements).

Each pixel produces light in combinations of red, green, and blue (RGB).

By mixing different amounts of these three colors, ANY color can be created!

Your screen refreshes 30-60 times per second - too fast for your eyes to notice!

Farnsworth's original idea used an electron beam scanning line by line (CRT technology).

SCIENCE FACT BOX

- * A 4K TV has over 8 million pixels, each with 3 sub-pixels (24+ million tiny lights!).
- * Farnsworth first demonstrated TV at age 21 with a simple straight line image.
- * He said he hoped TV would be used for education - 'There is nothing worthwhile on it.'

TRY THIS!

Look at a screen with a magnifying glass. Can you see the tiny red, green, and blue sub-pixels?

THINK ABOUT IT:

How did looking at rows in a potato field inspire the biggest invention of the century?

Chapter 3: Frank Epperson

Age 11 - Accidentally Invented the Popsicle

In 1905, 11-year-old Frank was mixing powdered soda and water on his porch. He accidentally left the cup outside overnight with the mixing stick still in it. That night, the temperature dropped below freezing in San Francisco. The next morning, Frank found his drink frozen solid around the stick! He pulled it out and had the world's first popsicle! He called it the 'Epsicle.' Years later, he patented it and sold millions. All from a happy accident!

THE SCIENCE: FREEZING AND SOLUTIONS

Pure water freezes at 32F (0C), but dissolved substances lower the freezing point! That is why Frank's sugary soda water froze at a lower temperature than plain water. This is called 'freezing point depression' - salt on icy roads uses the same principle! When water freezes, molecules slow down and form a crystal structure (ice). Adding sugar or salt disrupts crystal formation, requiring colder temperatures to freeze.

SCIENCE FACT BOX

- * Americans eat over 2 billion popsicles every year!
- * Ocean water freezes at about 28.4F (-2C) because of dissolved salt.
- * Ice cream trucks exist because Frank's 'accident' created a whole new industry!

TRY THIS!

Make popsicles! Freeze juice in cups with sticks. Try salt water vs sugar water - which freezes first?

THINK ABOUT IT:

Many great inventions were accidents. Why is being observant so important for scientists?

Chapter 4: Param Jaggi

Age 14 - Invented a CO2 Filter for Cars

At 14, Param Jaggi was concerned about car exhaust polluting the air. He knew that plants convert CO2 into oxygen through photosynthesis. 'What if I could make a device that does what plants do?' he wondered. He created a device that attaches to car exhaust pipes and uses algae to absorb CO2! The algae use the CO2 for photosynthesis, converting it to oxygen before it leaves. His invention won numerous science awards and inspired other young inventors.

THE SCIENCE: AIR PURIFICATION

Cars produce CO2 when they burn gasoline - a greenhouse gas that warms our planet. Photosynthesis: plants/algae use CO2 + sunlight + water to make sugar + oxygen. Algae are incredibly efficient at photosynthesis - even better than trees per volume! Carbon capture means removing CO2 from the air or exhaust before it enters atmosphere. Engineers are developing many ways to capture carbon: algae, minerals, and technology.

SCIENCE FACT BOX

- * A single car produces about 4.6 metric tons of CO2 per year!
- * Algae can double their population in just hours - incredibly fast growers.
- * Trees absorb CO2 but algae can be 10-50 times more efficient per unit area!

TRY THIS!

Grow algae! Put pond water in a clear jar in sunlight. In days, green algae will multiply!

THINK ABOUT IT:

What everyday problems around you could be solved with a creative invention?

Chapter 5: Gitanjali Rao

Age 11 - Created a Lead Detection Device

When 11-year-old Gitanjali saw news about the Flint, Michigan water crisis, she was shocked that families were drinking lead-contaminated water without knowing. 'Testing for lead is expensive and slow,' she learned. 'I want to change that!' She invented 'Tethys' - a device that detects lead in water quickly and cheaply. It uses carbon nanotube sensors that change when lead atoms attach to them. In 2017, she won the Discovery Education 3M Young Scientist Challenge!

THE SCIENCE: WATER CONTAMINATION

Lead is a toxic metal that can damage the brain, especially in children. Old lead pipes can leach lead into drinking water - invisible and tasteless! Standard lead tests take days and cost hundreds of dollars. Gitanjali's device: seconds! Carbon nanotubes are tiny tubes of carbon atoms - incredibly sensitive chemical sensors. The sensor changes electrical resistance when lead atoms bond to it - detectable by a phone!

SCIENCE FACT BOX

- * Even tiny amounts of lead (parts per billion) can harm developing brains.
- * Gitanjali was named TIME Magazine's first-ever 'Kid of the Year' in 2020!
- * She has since created apps to fight cyberbullying and detect other contaminants.

TRY THIS!

Research your local water quality report. What gets tested? Are there any concerns?

THINK ABOUT IT:

What problems in your community could technology help solve?

Chapter 6: Jack Andraka

Age 15 - Developed a Low-Cost Cancer Test

When a close family friend died of pancreatic cancer, 15-year-old Jack was devastated. He learned that the existing test was 60 years old, expensive, and missed 70% of cases! 'I can do better,' he decided. He spent months researching at the library and online. He developed a paper sensor that detects a cancer protein 168 times faster! His test costs just 3 cents (vs \$800!) and takes 5 minutes (vs hours!). After being rejected by 197 labs, one professor said yes - and Jack proved it worked!

THE SCIENCE: BIOSENSORS

Biosensors detect biological substances (proteins, DNA) using chemical reactions. Jack's sensor uses antibodies that bind specifically to a cancer protein (mesothelin). When the protein attaches to antibodies on the paper, electrical properties change. Carbon nanotubes on filter paper create a network that responds to the protein binding. This is similar to how COVID rapid tests work - antibodies detecting specific targets!

SCIENCE FACT BOX

- * Jack was rejected by 197 researchers before finding one who believed in his idea!
- * His test is 26,000 times less expensive and 400 times more sensitive than existing tests.
- * Persistence paid off - he never gave up despite nearly 200 rejections!

TRY THIS!

Research 'antibodies' - how do they recognize specific things? Draw how a lock-and-key system works.

THINK ABOUT IT:

Jack was rejected 197 times before succeeding. What does this teach about persistence?

Chapter 7: Kelvin Doe

Age 13 - Built Batteries and a Radio from Trash

In Sierra Leone, Africa, 13-year-old Kelvin had no access to electronics or electricity. But he was endlessly curious and loved taking things apart to see how they worked. He collected discarded electronics and scrap metal from trash dumps. From this 'junk,' he built batteries, a generator, and eventually a radio station! His community radio station broadcast music and news to his whole neighborhood. He became the youngest person ever invited to MIT's 'Visiting Practitioner' program!

THE SCIENCE: BATTERIES FROM RECYCLED MATERIALS

Kelvin made batteries by combining acid, soda, and metal from recycled electronics. A battery needs two different metals and an electrolyte (acid/base solution). Chemical reactions between the metals and electrolyte produce an electrical current. His generator used a motor in reverse: spinning it by hand produced electricity! Radio transmitters convert electrical signals into electromagnetic waves.

SCIENCE FACT BOX

- * Kelvin had no formal engineering training - he taught himself from discarded textbooks!
- * He created his own radio station: 'DJ Focus' broadcasting to his community.
- * He later started a company to bring power to communities without electricity.

TRY THIS!

Build a simple battery: put copper and zinc strips in lemon juice. Measure voltage with a multimeter!

THINK ABOUT IT:

How does Kelvin's story show that creativity is more important than resources?

Chapter 8: YOUR Turn!

The Next Great Inventor Could Be YOU!

Every inventor in this book was once just a kid with a question or a problem. They did not have special powers or expensive labs. They had CURIOSITY and PERSISTENCE. Louis was blind. Kelvin had no electricity. Jack was rejected 197 times. But they all shared one thing: they saw a problem and decided to solve it. The world has millions of problems still waiting for creative solutions. And the next great inventor might be reading this book right now. That means YOU!

THE SCIENCE: THE SCIENTIFIC METHOD

Step 1: OBSERVE - Notice a problem that needs solving.
Step 2: QUESTION - Ask 'Why does this happen?' or 'How can I fix this?'
Step 3: RESEARCH - Learn everything you can about the problem.
Step 4: HYPOTHESIZE - Make an educated guess about a solution.
Step 5: EXPERIMENT - Build a prototype and test your idea.
Step 6: ANALYZE - Did it work? What can you improve?
Step 7: SHARE - Tell others about your findings!

MORAL:

You are never too young to change the world with science!

SCIENCE FACT BOX

- * The average age of a first patent in the US is getting younger every decade!
- * Many of the world's biggest companies were started by teenagers in garages.
- * Every problem you notice is an opportunity for an invention!

MY INVENTION WORKSHEET

1. THE PROBLEM I want to solve:

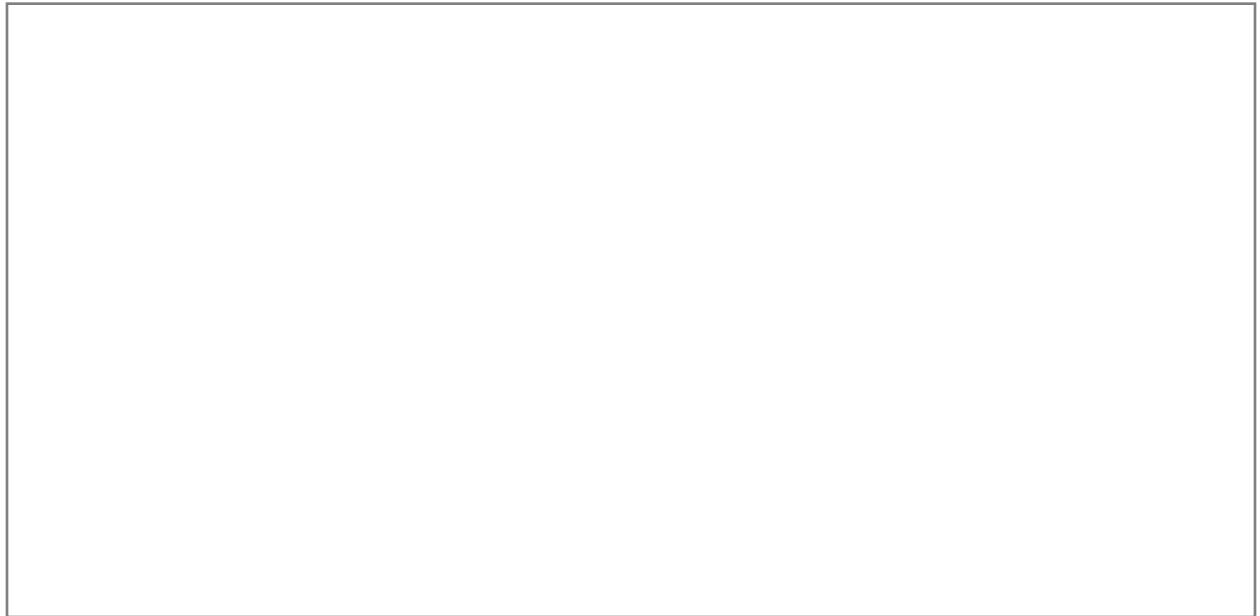
2. WHO has this problem?

3. MY SOLUTION idea:

4. MATERIALS I would need:

5. How would I **TEST** it?

6. DRAW your invention here:



INVENTOR'S QUIZ

1. What did Louis Braille invent and why?

Answer : _____

2. What inspired Philo Farnsworth's TV idea?

Answer : _____

3. How was the popsicle invented accidentally?

Answer : _____

4. How does Param Jaggi's CO2 filter work?

Answer : _____

5. What problem did Gitanjali Rao solve?

Answer : _____

6. How many times was Jack Andraka rejected before success?

Answer : _____

7. What did Kelvin Doe build from trash?

Answer : _____

8. List the steps of the scientific method.

Answer : _____

9. What do all these young inventors have in common?

Answer : _____

10. What problem would YOU like to solve?

Answer : _____

FUTURE INVENTOR CERTIFICATE

I am inspired to change the world!

The inventor who inspired me most: _____

My invention idea: _____

The problem it solves: _____

Name: _____

Date: _____

Future World-Changer!